The Promise of Augmented Reality in Everyday Life

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Augmented Reality (AR) is increasingly positioned as the next frontier in personal computing, yet how it would actually integrate into everyday life remains poorly understood. We investigate the promise of everyday AR by analyzing (1) 184 practitioner-created vision videos and (2) 1467 user-generated visions based on first-person recordings of 30 participants' everyday lives. From this dual perspective of both practitioner knowledge and lived experiences, we develop a conceptual framework that describes why users might seek to augment reality, what is missing from everyday experiences that AR could provide, which aspects of reality are useful to augment, and how AR content could relate to the real world. Our findings reveal new opportunities for everyday AR and provide a framework to think about AR use in diverse real-world settings.

CCS Concepts: • **Human-centered computing → HCI theory, concepts and models**; **Mixed / augmented reality**; *Empirical studies in HCI*; *Ubiquitous computing*.

Additional Key Words and Phrases: augmented reality, smart glasses, everyday life, context-awareness, mixed reality, spatial computing

1 Introduction

With the release of devices such as the Apple Vision Pro[1] and Snap Spectacles[2], augmented reality (AR) technologies are gaining renewed attention as personal computing devices. They are often envisioned to transform our everyday lives, much like how smartphones and desktop computers did in the past [1, 6]. To advance this vision, researchers have produced numerous designs and insights into AR user interfaces [43], tracking [53, 79], and context-awareness [33, 59].

While these efforts are crucial for imagining what novel AR interactions are possible and technically feasible, our current knowledge of how AR might integrate into everyday life remains limited. Prior research shows that usage depends not just on what is possible but on context, existing behaviors, values, and mental models [63, 73]. Thus, empirical work that investigates how users envision and appropriate AR in diverse, real-world contexts is required to complement technological advances. However, such work remains underexplored as constraints on

[1]https://www.apple.com/apple-vision-pro/
[2]https://www.spectacles.com/

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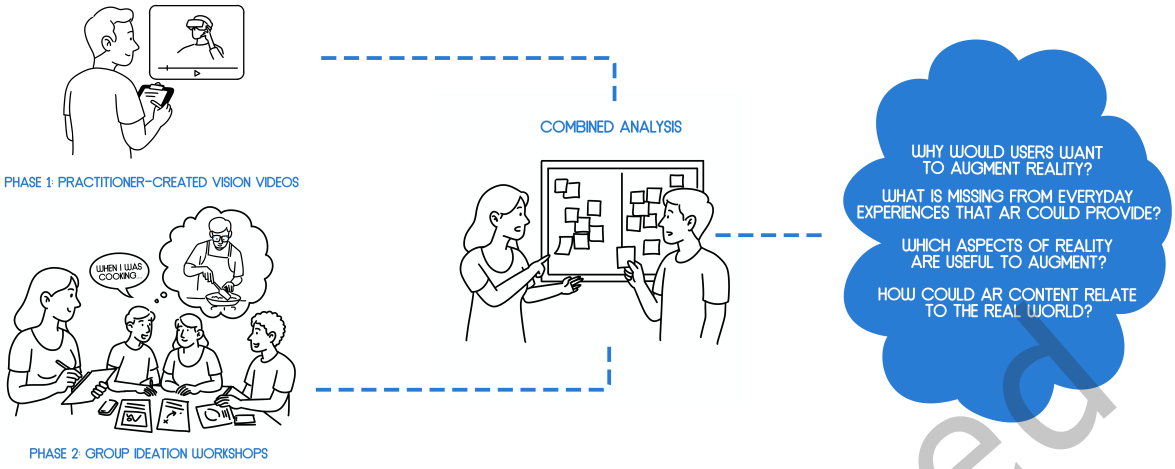

Fig. 1. A diagram of our process for constructing a conceptual framework for how AR could integrate into everyday life

hardware currently restrict studies on AR use to short-term interactions, often in lab settings, where only specific parts of reality are augmented. This contrasts starkly with how, for example, smartphones are used in everyday life [20, 21], where diverse, dynamic and spontaneous usage occurs in varied environments. As researchers, our understanding of how AR will integrate into everyday life is also constrained by existing paradigms [71, 92], known application domains [48, 96], and trends in the design of commercial hardware [15].

For these reasons, we do not know well how end users will appropriate AR in their everyday lives. While existing models and frameworks for AR enable creating, delivering, and interacting with AR content, they do not discuss how users could merge AR with doing the laundry, engaging with other people, or cooking a celebratory meal. Grubert et al. [37]'s taxonomy lists goals and attitudes as an important context source for pervasive AR systems but do not further elaborate what these could be. Zollmann et al. [104]'s taxonomy of visualization techniques for AR explains different ways of displaying content but not what types of content are valuable for such everyday uses and which aspects of everyday life could benefit from augmentation. Similarly, there are models of how AR content can relate to parts of the real world, such as being situated near them [54] or hiding them [27], but we lack a unified understanding of what these relationships could be. To bridge these gaps, this work looks into potential AR futures and investigates what AR *could become* as it advances and expands.

To do this, we analyze a large number of visions of what everyday AR use could look like from two complementary sources: (1) 184 practitioner-created vision videos and (2) 1467 user-generated visions based on first-person recordings from 30 participants' everyday lives. Previous works [32, 49] have argued that such visions are powerful drivers of innovation and thus have practical implications for developing new technologies. Thus, we begin by analyzing what practitioners imagine AR could do, drawing on creative depictions of AR use cases from companies, artists, and developers, which we refer to as vision videos for this work. We then analyze user-generated visions, derived from workshops where participants brought first-person recordings of their everyday lives and envisioned how AR could enhance those moments. While the vision videos are made by those with experience with AR, giving a technology-focused view of AR, the user-generated visions from the workshops give a more pragmatic view based on individuals' ideas about augmenting reality. This contrasts positive and polished views of AR with more mundane or even chaotic views of AR. The two data sources also balance breadth and depth, helping us better understand how AR might fit with varied environments, tasks, and needs. The vision videos incorporate use cases from multiple regions and the user-generated visions provide

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situated, fine-grained glimpses into everyday activities of a locally recruited but demographically heterogeneous group. By combining insights from both sources, we contribute the "Promise of Augmented Reality in Everyday Life" that explains:

(1) Why would users want to augment reality?
(2) What is missing from everyday experiences that AR could provide?
(3) Which aspects of reality are useful to augment?
(4) How could AR content relate to the real world?

The answers to these questions constitute a conceptual framework of how AR integrates with everyday life, providing a structured lens for researchers and practitioners to understand and ideate about everyday AR use. Further, this framework can guide everyday AR systems in supporting users in contextually appropriate and meaningful ways, as well as help users better communicate their requirements and expectations to these systems.

2 Related Work

2.1 Applications of AR to Everyday Life

The idea of AR being a continuous part of our lives has been implicit in many visions of AR for decades. Grubert et al. [37] laid out one concrete instance of this vision, which they coined "Pervasive Augmented Reality" to describe AR systems that can adapt to changing requirements and constraints of a user's context. Extending this idea, Lu and Bowman [61] proposed the paradigm of "Glanceable AR" where information is present in a user's view such a way that it can be accessed by a quick glance when needed but stays unobtrusive otherwise. Further research on this paradigm [25, 29, 62] has explored glanceable AR content in social and on the go settings. These works have identified relevant context targets and content placement strategies for everyday AR, but have left unanswered the question of which parts of reality are most relevant to augment in our everyday lives and why.

Existing research has also demonstrated AR's potential in various everyday contexts. For instance, Mathis et al. [67] proposed using virtual replicas of real-world objects to preview and manipulate near-future scenarios that align with the current context. Bhatia et al. [16] developed AR annotations inspired by comic books to represent non-visual sensations and studied if such effects could alter perception for everyday activities such as eating and running. Extending outside the visual domain, researchers have explored haptic augmented reality [14], which changes how it feels to touch objects rather than just see them in everyday tasks. Additionally, previous works have examined AR for specific tasks, such as using it in the kitchen [9, 83], in the gym [36], and for performing calculations [81].

These works are exciting examples of what AR can bring into the world, but they are isolated, task-specific applications. They do not capture the breadth of activities people do in their everyday lives and restrict AR use to artificial settings rather than peoples' own tasks, belongings, and environments. Thus, the broader question of the promise of AR in everyday life remains unexplored.

2.2 Studies on AR Use in Everyday Life

Researchers studying the use of AR in everyday life typically analyze how existing AR hardware is employed by people in their lives. Orchard et al. [77] focused on how users employ the Focals by North smart glasses to facilitate everyday activities. They conducted a focus group to discuss experiences and invited participants to brainstorm future use cases. However, these use cases were related to existing software available on the glasses rather than being anchored in moments from everyday life and were not analyzed further. Woodward et al. [103] conducted participatory design sessions with children to generate ideas for AR use, which they then used to construct a mental model for how children understand AR headsets. Cheng et al. [26] and Pavanatto et al. [80] have conducted studies on how people perform knowledge work with AR displays instead of regular laptops. Kim et al. [46] conducted a field study to understand the experience of performing various everyday activities

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while wearing a pass-through AR headset. Their findings offer valuable design insights for AR hardware and software aimed at everyday use, which are relevant to the use cases discussed in this paper. Tran et al. [97] analyzed YouTube videos uploaded by users of existing AR headsets and smart glasses to understand compelling use cases and challenges for these devices.

While these studies contribute to understanding AR's role in everyday life and offer insights into current technologies, they are largely constrained by the limitations of existing hardware. Moreover, they often emphasize conventional screen-based use cases, such as media consumption and communication, rather than exploring novel possibilities uniquely enabled by AR. In contrast, our work moves beyond current capabilities to explore the broader promise of AR in everyday life, unconstrained by today's technological limitations and available applications.

Some researchers have discussed the future potential of AR use for everyday life but do so in a narrower scope than our work. Knierim et al. [50] discussed opportunities for using AR at home by conducting an online survey and user study. In their study, participants took pictures inside their homes to illustrate future AR scenarios. Our work extends this by using first-person video to capture richer context and by exploring AR use across a wider range of everyday environments beyond just the home. Mathis [66] discussed AR as a technology that helps overcome challenges in daily life such as recalling information or comprehending information. We build on this work by considering motivations beside challenges, including hedonic, expressive, and enhancement purposes. Along with benefits of AR in everyday life, researchers have also highlighted potential negative consequences, such as differences in perceived realities [78] and using filters to modify the appearance of others or ourselves in everyday scenarios [18]. Our work complements these perspectives by offering a grounded yet imaginative exploration of both the mundane and utopian roles AR could play in everyday life.

Overall, while prior works have provided valuable insights into current and near-future AR use, they are limited by technological constraints and not broad enough to capture AR use across diverse contexts. By analyzing both practitioner visions and user-generated ones, we move beyond these limitations and construct a boarder understanding of how AR integrates into everyday life.

2.3 Descriptive Models for Augmented Reality

Descriptive models are used to categorize, explain, or structure a problem space [64]. In human-computer interaction research, this can take place in the forms such as design spaces [8, 13], frameworks [70, 87], and taxonomies [23, 57]. Such descriptive models have been proposed for AR as well, which Normand et al. [75] have classified into four categories: technique-centered, user-centered, information-centered, and interaction-centered. The aim of our work is to go further than these categories by creating a framework that is not only focused on the user or the AR system, but on how they interact with reality.

Technology-centered models focus on the technical components of AR systems such as the display or sensors used. Zollmann et al. [104] created a taxonomy of visualization techniques in AR in terms of the goal of the visualization, visibility of data, depth cues, filtering or abstraction of data, and compositing methods. Due to our focus on visions to explore beyond technical limitations, our framework is technology-agnostic. Thus, we complement such technical models by describing what content to display while they describe ways of displaying content.

User-centered taxonomies focus on how AR affects the human sensory system and perception. Lindeman and Noma [58] described AR systems in terms of the stimulus pathway used for mixing the real world and computer generated stimuli together. Langlotz et al. [54] presented a design space for vision augmentations in terms of the source and target visual domains, such as leveraging binocular vision to augment photo perception. In contrast to such models, we focus on the user's lived experiences and intentions, offering a purpose-driven perspective rather than one based on perception.

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Information-centered models focus on the information available to the AR system and how to present it. Such models have been proposed by the visualization community to describe ways to embed data into the spaces, objects, and entities to which it corresponds [101]. Lee et al. [56] cataloged ten design patterns for such in-situ data visualization using AR. Our work aims to extend such information-centered models by combining the notion of embedded and situated visualization techniques with concepts from mediated reality [27, 90] to offer a richer understanding of how AR content can relate to and transform the real world.

Interaction centered models describe how users interact with AR content. Hertel et al. [43] proposed a taxonomy of immersive AR interaction techniques in terms of two dimensions: user tasks and input modalities. Interaction techniques are out of scope for our work as we focus on application areas and goals rather than how the user interacts with virtual content in terms of selection or manipulation.

A category of AR taxonomies that our work can be classified into is application-centered ones such as those for collaboration [65], augmenting zoo exhibits [95], and education [72], with our domain being the broad and underexplored area of everyday life. Overall, our aim is to complement existing descriptive models for AR by shifting the focus from how AR works to what AR should do and why it matters. By centering on a diverse and expansive set of everyday use cases, our framework offers a new lens for understanding AR's potential and guiding its integration into daily life.

3 Methodology

To capture a broad set of use cases, we combined two established methods: analyzing videos from video sharing platforms [44, 51, 69] to understand practitioner visions for AR use, and recording and analyzing real-world interactions to examine how people engage with technology in everyday contexts [22, 82]. The latter, in our case, focuses on how participants would like to use AR rather than capturing how it is used. This approach combines top-down visions of technological potential with bottom-up expressions of user needs, providing a holistic view of the future of AR.

We scope our work based on the definition of visual AR by Azuma et al. [7]. Since many works use AR and MR interchangeably and the distinction between the concepts is not universally agreed upon [92], we also used mixed reality as a keyword in our video search. Our use of the term "everyday life" in this work refers to the mundane, habitual, and routine activities captured in our two data sources. It is not meant to convey all people's

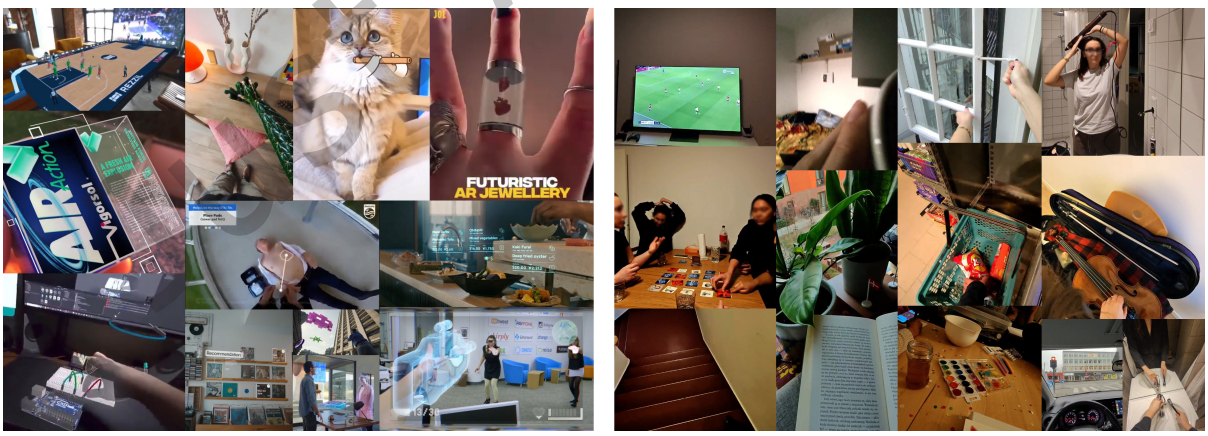

Fig. 2. A snapshot of the two sources of data we analyzed: practitioner-created visions videos (left) and user-generated visions of AR use based on first-person recordings of people's everyday lives (right)

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life experiences across global demographics and life stages. We note that scoping our analysis to AR content that relates to the real world reflects the focus of this work and not the full range of ways AR will be used. AR displays and smart glasses can of course show information unrelated to the user's immediate surroundings, such as notifications or documents, and these can be compelling reasons for using AR technology along with augmentations directly tied to reality.

3.1 Phase 1: Practitioner-created AR Vision Videos

In the first phase, we collected 184 vision videos of future interactions with AR. We collected videos through a snowball sample, as in similar studies [5, 44], and analyzed their content qualitatively (see Figure 3 for an overview of the process).

3.1.1 Collecting Videos. To identify appropriate search terms, we first collected 46 videos, including advertisements from major manufacturers of AR technology, clips from science fiction media, and other videos that the authors, who are AR researchers, were aware of. We then selected seven keywords that appeared most frequently across the collected videos and captured our objective to focus on AR: "augmented reality", "mixed reality", "XR", "metaverse", "smart glasses", "digital overlay", and "holographic".

We chose Youtube, LinkedIn and X(Twitter) as the platforms for our search, because they are the largest platforms in terms of users and content, supporting both short and long format videos. We also considered TikTok and Instagram, but the relevant videos we found on these platforms were already covered by LinkedIn and X.

For Linkedin and X, we created new accounts with no connections and accounts being followed to ensure that the search is not optimized for links to existing accounts[3]. We then performed the search on the 25th and 26th of May 2024 by running a separate search for each term. The first 50 video-posts for each search term and platform were screened. For Youtube, we also created a new account and watched all Youtube videos from our initial set to prime the algorithm with the videos we were interested in. This was done as the search on Youtube is also customized based on the relevance to the user [4]. Since Youtube also recommends related videos, we looked at all the recommendations for the videos, which were marked as relevant. For all three platforms, we used the default sorting of search results, which is relevance.

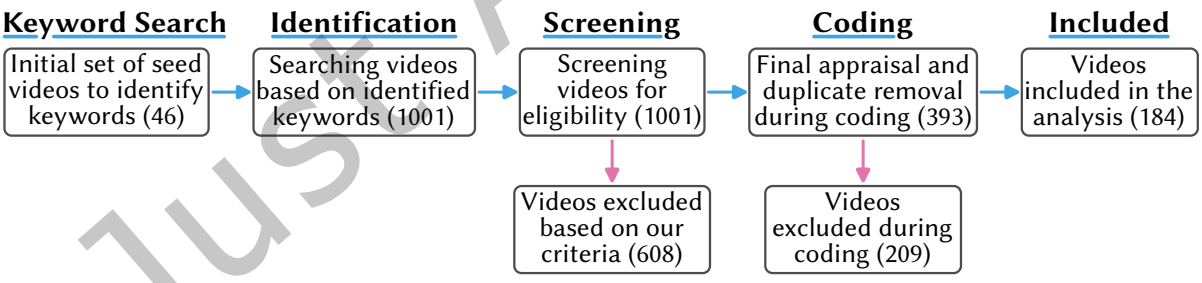

Fig. 3. Process for collecting AR vision videos

3.1.2 Screening. To identify relevant videos, we developed the following criteria through watching and discussing the seed videos among three of the authors.

[3]https://www.linkedin.com/help/linkedin/answer/a524188, https://help.twitter.com/en/using-x/top-search-results-faqs
[4]https://www.youtube.com/howyoutubeworks/product-features/search/

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- The video must show both digital content and the real world or an unaltered proxy of the real world (in the case of animated videos) in the same frame. Purely digital content was hence excluded (e.g., virtual reality).
- Since we are interested in the combination of everyday life and AR, the digital content must be related to the real world or user context in some way (e.g., registered in physical space, conveying information about physical objects). Hence, purely screen-based use cases such as watching a video floating in the air were excluded.
- The digital overlay is targeted at what a user or a person in the video is perceiving. We excluded videos consisting exclusively of annotations meant to aid the understanding of viewers of the videos (e.g., subtitles).
- If the user interacts with the digital content as they would with the physical counterpart of that content, they must acknowledge the digital nature of it in the video (this helps exclude special effects from movies).

One author watched and marked all the videos for inclusion in a spreadsheet and removed duplicates across platforms. Different videos that conveyed the exact same concept were treated as duplicates. Even if only a small portion of the video was about AR, it was still included but only the AR portion of the video was marked for analysis. This resulted in 184 videos included for analysis. The set of videos consisted of concept videos from companies, clips from TV shows or movies, demos made by hobbyists, and recordings of AR applications being used. No videos corresponding to research papers were collected, likely due to them not being favored by the search algorithm as they are unlikely to be of interest to non researchers. The links to the videos can be found in the supplementary materials.

3.2 Phase 2: Generating Visions Anchored in Everyday Situations Through Workshops

In the second phase, we aimed to understand how AR might be used in scenarios that are more mundane than the ones in commercially produced vision videos. To capture such scenarios, we used in-situ recordings of events, inspired by experience sampling [55]. As we outline in Figure 5, our participants worked with us both to record aspects of their everyday lives, as well as to examine and discuss their own and others experiences in ideation workshops. As Rosner puts it, workshops are a way by which researchers can surface "common sense understandings of ongoing activity [that] render actions observable and reportable" [86]. That is, a way of foregrounding easily forgotten everyday experiences and turning them into ideas.

3.2.1 Participants. We recruited thirty participants using social media posts and advertisements at local universities. Our participants were between the ages of 21 and 39, with 19 identifying as female and 11 as male. While recruitment was localized to a single metropolitan region, the participants varied in nationality (13 countries), occupation (students, office, public service, creative, remote work, unemployed), and living arrangements (alone, with a partner or family, shared apartments, student dorms). Further detail on participant demographics can be found in the appendix A.

3.2.2 Procedure. Participants were instructed to use a pair of Ray-Ban Meta smart glasses [5] to record slices of their everyday life. These glasses do not feature an AR display, but have an inbuilt camera which can record short first-person videos (video and sound) lasting up to 3 minutes. We asked the participants to wear or have the glasses available as much as possible for two consecutive days. To ensure a broad sample of recordings at random moments, we also asked them to record whenever we sent a notification through SMS, which we did eight times each day. We instructed participants to record as soon as they felt comfortable recording after receiving a notification. The notifications were sent based on the participants' waking hours, which we divided into eight segments with one reminder being sent at a random time in each segment. To help maintain privacy

[5] https://www.ray-ban.com/usa/ray-ban-meta-ai-glasses

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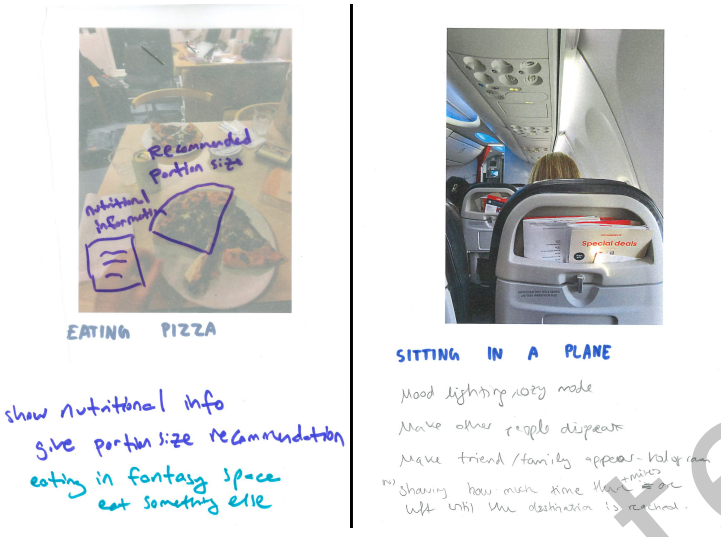

Fig. 4. Examples of ideas generated by the participants based on their recordings

and minimize the burden of participation, we emphasized that the decision of whether to record was theirs. We also explained how to delete recordings and allowed participants not to share any of their videos.

Within ideation workshops, participants worked together in groups of three on sharing, recounting, reflecting, and ideating on their recordings and those of others. During the workshops, we used the word "magic glasses'' (and not "smart glasses'' or "AR glasses'') to reduce biases based on their understanding of these concepts or technological possibilities and to provide scaffolding to the ideation process as recommended by Andersen and Wakkary [4]. We printed the first frame of each video recorded by the participants on an A5 sheet of paper, where the participants could write down ideas for using magic glasses. The number of videos recorded by each participant ranged from 10 to 43. If a participant recorded more than 15 videos, the researcher running the workshop excluded the videos corresponding to moments similar to those captured by other participants, ensuring a broad mix of situations. The participants were encouraged to communicate their ideas for each video in any way they felt comfortable, whether writing down or drawing on the printouts. They were allowed to skip videos if they could not come up with any ideas for them. Workshop sessions took an average of 75 minutes and were audio-recorded and transcribed for further analysis.

The workshops were structured into three ideation rounds: First, the participants individually noted how they would use magic glasses in each recorded video for 5 minutes, followed by 10 minutes of group sharing and discussion. In the second round, participants passed their phone and printouts to the person on their right, spent 5 minutes generating ideas for someone else's videos, and then engaged in another 10-minute group discussion. This process was repeated, so everyone contributed to the other two participants' videos. In the final round, participants returned to their own videos, drew from a deck of eight prompt cards inspired by Oblique Strategies [34] (see supplementary materials), and generated new ideas based on the prompts for 5 minutes. The workshop concluded with a final group discussion on the prompt-based ideas.

3.3 Analysis

We first individually analyzed the two sources of data and then integrated the insights from both to construct our final results. Our analysis is qualitative, following a bottom-up, inductive coding process, clustering open

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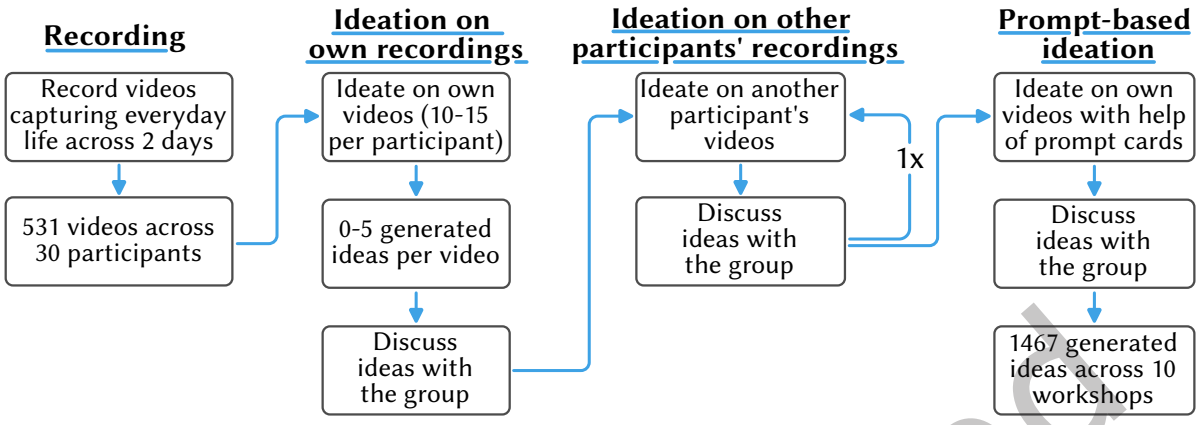

Fig. 5. Process for generating visions anchored in recordings of everyday life

codes into themes through affinity diagramming [40, 45] to cover the range of possibilities for everyday AR. Both phases of our study are intended to capture a large and diverse set of random everyday activities rather than a specific uniformly distributed set, hence, we do not report how frequently specific applications are mentioned [88].

3.3.1 Phase 1 Analysis. To analyze the vision videos, three of the authors watched a random set of five videos together and made notes about what was noteworthy in them regarding the integration of AR and everyday life that our analysis should capture. The notes were then compared and discussed to develop an initial coding scheme. Ten new videos were individually coded using this scheme by the same authors, and the scheme was discussed in a group to resolve differences and refine it. The coding scheme was then discussed with a fourth author who provided additional feedback resulting in the final scheme consisting of the following four components:

- **Why:** Why is AR suited for this specific situation?
- **What:** What content is brought into the real world by the AR system?
- **Which:** Which entity in the real world is AR operating on?
- **How:** How do AR content and the real world relate to each other?

The set of ten initial videos was re-coded using this final scheme and the rest of the videos were distributed equally among three authors. We then applied the coding scheme to each instance of AR content in a video and hence some videos led to multiple sets of codes. As AR and HCI researchers, our experience with prior works in mediated reality [27, 90] and visualization [54, 56, 101] informed the terminology used during coding. The *What*, *Which*, and *How* directly capture the contents of the videos but the *Whys* required some level of interpretation (e.g. in the case of AR task guidance, coding it as either "helps with understanding" vs "provides access to information"). Hence after all videos were coded, one of the coders went through all the *Whys* to double check and refine the interpretation of suitability of AR in that scenario. After coding, we used Miro [6] for affinity diagramming, importing each code as a sticky note. Four of the authors then met and first discussed about twenty notes for each component to create an initial set of clusters. We then independently and in parallel clustered the remaining codes for ten minutes. After ten minutes, there was a pause and the clusters created until that point were discussed and refined. The process of independently clustering and discussing was repeated across two sessions till all the codes were assigned to a cluster.

[6]https://miro.com/

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3.3.2 Phase 2 Analysis. The collection of ideas resulting from the workshops consisted of ideas collected by the participants themselves in the form of written idea notes along with ideas 'in flight' that emerged from participants' discussions. We sought to be as liberal as possible in generating the number of ideas, so as to capture different variations on the ideas as well. The workshops thus produced a large corpus of ideas, with 1467 ideas generated in total. We define an idea as a concept, solution, or thought expressed verbally, in writing, or drawing by a participant during the workshop that describes a way for magic glasses to be used. We captured each idea by noting the idea itself, the situation it was created for, and any further comments by the participant who created it. The list of ideas generated can be found in the supplementary materials.

For analyzing the workshop data, the authors discussed a random subset of ideas to update the previous coding scheme for the phase 2 data. In the end, we decided to use the same scheme that was used to analyze the vision videos excluding the *How* code. The ideas generated in the workshops were obviously less detailed than the vision videos and thus often lacked visual descriptions preventing us to code for *How*. For example, P24 mentioned wanting to see random facts while taking the lift, but did not specify where they should be displayed, leaving open whether this information was hovering in the air, integrated with the elevator walls, or shown in some other way. Thus, the generated ideas were coded by one author focusing on the *What*, *Which*, and *Why* codes. To cluster the ideas, we followed a similar process as for phase 1, with each code being imported into a second Miro board as a sticky note followed by rounds of discussion and independent clustering. Four of the authors met across three sessions to affinity diagram the three sets of codes. Duplicate ideas across workshops were removed during the coding phase as well as during affinity diagramming.

3.3.3 Merging Findings. The analysis of data from two sources led to two separate affinity diagrams for each component except *How*, which only had one diagram corresponding to phase 1. While some clusters overlapped, many were distinct due to differences in the nature of the data and the contexts in which the discussions took place. To connect insights across both phases, we refined and restructured the clusters from each diagram, combining them to form a unified affinity diagram whose clusters form the components of the framework of "The Promise of Augmented Reality in Everyday Life." To do this, the first author conducted an iterative comparison of clusters across the two diagrams, merging similar ones, splitting larger clusters, and resolving discrepancies through discussions with two other authors. The resulting unified affinity diagram was then further reviewed and refined collaboratively.

4 The Promise of Augmented Reality in Everyday Life

Based on our empirical analysis of AR vision videos and participant visions that are based on recordings of people's everyday lives, we developed a conceptual framework of "The Promise of Augmented Reality in Everyday Life". This framework describes how AR could integrate into everyday life and consists of four components corresponding to our central questions: (1) Why would users want to augment reality, (2) What is missing from everyday experiences that AR could provide (3) Which aspects of reality are useful to augment and (4) How could AR content relate to the real world. Below, we describe the core components of the framework and further elaborate on them through additional details present in Tables 1, 2, and 3 along with Figure 10. When describing the framework, we use the notation Px, Lx, Tx, and Yx to refer to visions from participants, LinkedIn, X, and YouTube videos respectively.

4.1 Why?

This component of the framework describes the reasons for using AR to augment our everyday lives. These reasons highlight the advantages of AR systems over traditional screen-based devices such as smartphones and personal computers. Our analysis identified four categories into which reasons for AR use may fit that we describe below along with 24 sub categories within them (Table 1).

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4.1.1 Access. AR can provide access to people, places, objects, and information that are otherwise unavailable or difficult to reach. It bridges physical and conceptual gaps, allowing users to interact with things outside their immediate environment. For example, AR can offer **access to information in situ**, such as displaying game rules directly on the board while playing (P24), or provide **access to people who are not there**, like a distant friend acting as a virtual gym buddy (P21). It can allow **access to objects that cannot exist**, such as gravity-defying Christmas lights that can be arranged in any pattern (L139), or offer **access to physical objects one doesn't have**, like a virtual Arduino and sensors for learning prototyping (T52). Beyond objects, AR can simulate environments. It allows **access to physical spaces one doesn't have**, for instance, by making the view outside a window appear as if underwater (T42), or provide **access to other times**, such as visualizing how a monument looked when it was first constructed (Y15).

4.1.2 Cognitive Assistance. AR can support perception, thinking, and decision-making by enhancing how users interpret and interact with their surroundings. It can **aid sensory perception**, such as enabling night vision (P1), and **process elements in view** to help users interpret what they see, like identifying whether an item needs repair or replacement (P19). AR can be used to **track elements in view** to know which clothes owned by a user were worn last week (P27) and support **contextual AI use** to suggest outfits out of the ones that are still clean (P9). It can even allow **view-based automation**, such as adding detergent to a shopping list when supplies run low (P6). Through **deictic** cues to highlight relevant items, such as the other sock pair when folding laundry (P6), AR speeds up actions by reducing cognitive effort and enhancing situational awareness.

4.1.3 Enhancements. AR can enhance the real world by adding new layers of functionality, aesthetics, or interactivity. For instance, AR can be **cosmetic**, such as adding virtual bottles that make a bar look well stocked (Y56), or be used for **entertainment and gamification**, like awarding points for chopping vegetables (Y164). It can even **embellish** real-world experiences, such as adding a celebratory glow around the winner of a board game when they make the final move (P20). Beyond aesthetics and play, AR can **enhance physical objects**, for example, by adding a virtual forgot password option to a bike lock (P19). AR also enables content transformation, allowing users to **make static content dynamic**, such as animated tattoos (P25), or **make 2D content 3D**, like newspaper ads that pop out of the page (T59).

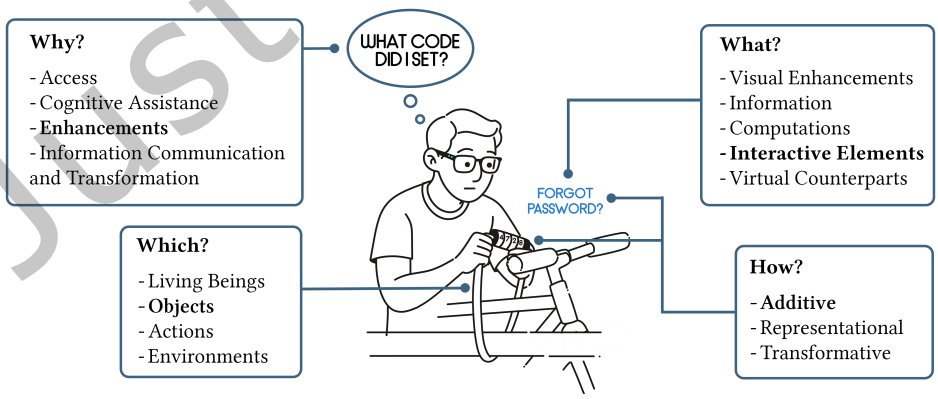

Fig. 6. An overview of the framework added to the "Add a forgot password option to a physical bike lock" scenario envisioned by P19

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Table 1. Why would users want to augment reality?

Why	Description	Example
Access		
Access to information in situ	Viewing relevant information directly in the situation where it is needed	Rules for a boardgame appear while playing it (P24)
Access to people who are not there	Seeing and interacting with virtual representations of people who you are not co-located with	Have a gym buddy even though you are not colocated (P21)
Access to objects that cannot exist	Bringing virtual objects that cannot physically exist into the real world	Christmas lights that don't respond to gravity (L139)
Access to physical objects one doesn't have	Bringing virtual objects into the real world that the user lacks physical access to	Prototyping using virtual Arduino and sensors (T52)
Access to physical spaces one doesn't have	Simulating being present in places where the user lacks physical access to	Make the view outside the window appear as if you are underwater (T42)
Access to other times	Observing past or future states and events in the location they take place	See how a monument looked like when it had just been constructed (Y15)
Cognitive Assistance		
Aid Perception	Improvements to sensory perception	Night vision mode (P1)
Deictic	Directly showing or pointing out things in the real world	Point to the other pair of sock when folding laundry (P6)
Contextual AI Use	Giving physical information and context as input to AI assisstants	Get suggestions for outfits to create when looking at wardrobe (P9)
Process Elements in View	Analyzing and interpreting elements of the real world	Know if something needs repair or has to be discarded or replaced (P19)
Tracking Elements in View	Identifying and monitoring activities based on what the user sees through the AR system	Wardrobe tracking to understand which clothes are clean or dirty (P27)
View based automation	Triggering actions based on what the user sees through the AR system	Automatically add items running low to shopping basket/list (P6)
Enhancements		
Cosmetics	Applying changes to physical appearances	Face filters (Y56)
Entertainment and Gamification	Enhancing physical tasks through the addition of virtual game elements	Turn chopping into a game with points based on how well you chop (Y164)
Enhancing Physical Objects	Adding additional functionality to real world objects through virtual elements	Add a forget password option for physical numerical bike lock (P19)
Embelishment	Adding additional decorative or non essential elements to objects and actions	Add glow around winner of boardgame (P20)
Make 2D Content 3D	Converting flat content into three-dimensional forms	Print advertisments with 3D effects (T59)
Make Static Content Dynamic	Allowing still images and objects to move, change or update	Dynamic moving tattoos (P25)
Information Transformation and Communication		
Transform physical information into another form	Converting real world data into a different format.	Turn physical book into an audiobook (P3)
Make information persistent in view	Making information visible and accessible at all times	See a checklist in view while packing (P17)
Personalize Information in view	Customize physical information to only to see what the user prefers to see	Convert prices to your own currency (P5)
Move physical information to where convenient	Relocating physical information to more accessible or useful positions.	Move expiry date from back of package to the front (P1)
Leave messages in the real world	Adding notes and messages to physical locations	Add a note about dangerous species in nature (P4)
Preview in situ	Visualizing how changes will look like in intended locations	Preview hairstyles before getting a haircut (P17)

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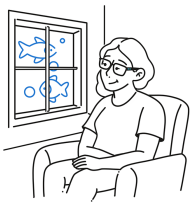
Point to other sock in the pair while folding laundry (P6)

Why: Diectic
What: Highlights
Which: Clothing
How: Embedded adding



Make view outside window seem like you are underwater (T42)

Why: Access to physical spaces one doesn't have
What: Nature
Which: Room
How: Replacing

Fig. 7. Examples of ideas from from P6 and T42

4.1.4 Information Transformation and Communication. AR enables new ways of interacting with, transforming, and communicating information to improve clarity, accessibility, and expressiveness. Rather than simply displaying data, AR can reshape how information is experienced in the real world. For instance, it can **transform physical information into another form**, such as turning a physical book into an audiobook (P3), or **make information persistent**, like keeping a checklist visible in the corner of your view while packing (P17). AR also supports personalization and spatial flexibility. Users can **personalize information in their view**, such as always seeing prices in their home currency while traveling (P5), or **move physical information where convenient**, like repositioning expiry dates to the front of packages for easier access (P1). Other than transformation and relocation, AR enables users to embed information directly into the environment. They can **leave messages in the real world**, such as placing a virtual warning sign on a dangerous plant (P4), or **preview changes in situ**, like visualizing different hairstyles before a haircut (P17).

4.2 What?

This section describes the virtual components that are brought into the real world by an AR system. These components are the building blocks of AR experiences that address gaps in the real world that the user faces. While the previous *Why* section described reasons for using AR, this section focuses on the AR content itself. We identified 27 types of content (Table 2) that the framework organizes into five different categories:

4.2.1 Visual Enhancements. are changes to the appearance of real-world elements to improve perception, aesthetics, or clarity. They emphasize, transform, or supplement visual information in ways that align with or extend the capabilities of the human visual system, such as highlighting a user's shoes in a shoe rack (P22), changing the color of the road in response to traffic lights (Y161), or seeing in the infrared spectrum (Y171).

4.2.2 Information Elements. are knowledge, data, or guidance that is overlaid onto the real world. They can be used to provide instructions to sort trash correctly (P18), satisfy the user's curiosity, such as the name of a plant they come across (P4), and even give explanations such as which door each key in a bunch opens(Y167).

4.2.3 Computations. are the results of digital processes that analyze, transform, or generate real-world data. AR can be used to compute information, such as the final price of a meal based on ingredients used (P24), generate suggestions, such as which combination of available ingredients would make the tastiest sandwich (P3), and even simulate how a room looked five months ago (P17). Unlike the previous category of information, computations involve some transformation of physical or other available data, not just querying it.

4.2.4 Interactive Elements. are digital components that enable users to actively interact with AR content, real-world environments, or other systems. These could range from simple UIs such as a virtual doorbell (Y161) to gamifying vacuuming a room into a coin-collecting game (L119), and even full-fledged virtual agents like an AI host to entertain people at a party (P12).

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Table 2. What is missing in the physical world that AR could bring to it?

What	Description	Example
Visual Enhancements		
Highlights	Emphasizing specific elements within the visual field	See a heart above friend in a crowd (P24)
Textures and Colours	Changing the appearance of surfaces in the real world	Virtual layer of paint on a wall (L114)
Aesthetic Elements	Elements that make the appearance of entities in the real world more attractive or pleasant	Make user look like Shrek to others (P15)
Perceptual Enhancements	Elements that help bypass existing limitations of the human visual system	Zoom into what you are seeing (Y1771)
Information		
Names	Displaying the words that an object or being is known by	See name of plants a user comes across (P4)
Status and Metrics	Information and measurements regarding the condition of aspects of the real world	Final price of meal based on ingredients used (P24)
Explanations	Clarifying aspects of the real world	Labels on keys explaining which doors each key opens (Y167)
Instructions	Guidance on performing tasks in the real world	Tempo visualization for performing chest compressions (L25)
Advertisments	Presenting promotional content in the world	Perimeter advertisements in a sports game customized to the user (L37)
Reminders	Notifications based on real world events or data	Reminder to close the window when leaving the house (P25)
Other Information	Information distinct from the previously mentioned kinds of information	See random facts while taking a lift (P24)
Computations		
Predictions	Forecasts about future events or outcomes in the real world	Estimated waiting time in line (P5)
Suggestions	Recommendations or advice based on physical data	Nail paint color suggestion that best fits skin tone (P23)
Data Visualization	Elements that encode data to enhance understanding and interaction	Visualization of number of trees killed till now due to printing (P7)
Simulation	Virtual models that demonstration possible states of real-world systems or processes	Flood water inside a room to understand effects of climate change (L3)
Recordings	Captures of real world data that can be subsequently played again	Relive memory of when parents came to visit (P17)
Analysis and Processing	Examination and manipulation of physical information	Identify spots which are not clean (P9)
Checks and Alerts	Verifications regarding physical actions and data	Alert on drinking tapwater in foreign country if it is unsafe (P7)
Interactive Elements		
UIs	Virtual mediums for communication between the user and systems in the real world	Virtual thermostat controls (Y129)
Games and Game Elements	Elements that allow play in the real world or make real world events playful	Empyting dishwasher seems like unearthing treasure (P6)
Virtual Agents	Virtual AI entities that interact with users or systems in the real world	AI referee for badminton game (P12)
Virtual Counterparts		
Clothes	Virtual garments or accessories	Funny hat for a cat (P28)
Lighting Elements	Virtual devices or systems that can illuminate spaces	Virtual lighting based on the user's mood (P21)
Print Media	Virtual objects corresponding to publications	Greeting card with 3D effects (L50)
Nature	Components of the natural world	Tropical birds in a city garden (P28)
People	Human beings that exist or existed in the real world	Gordon Ramsey commenting on food (P6)
Other Entities	Virtual entities distinct from the previously mentioned entities	Cornhole game elements (L112)

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Turn vacuuming into a coin collection game (L119)

Why: Entertainment and Gamification
What: Games and Game Elements
Which: Tool
How: Embedded adding

AI host for a party (P12)

Why: Access to people who are not there
What: Virtual agents
Which: Event
How: Embedded Adding, Replacing

Fig. 8. Examples of ideas from L119 and P12

4.2.5 Virtual Counterparts. are digital representations of physical or conceptual entities that exist in the real world. They simulate the presence, appearance, or behavior of real-world objects, beings, or phenomena, thus enabling users to perceive or interact with them as if they were physically present. AR can mimic objects such as jewelry (T7), tools like paintbrushes that can apply virtual paint to physical walls (T56), and even people, such as an AI copy of Anthony Bourdain who explains the culture of a place (P19).

Final price of the meal based on ingredients used (P24)

Why: Process Elements in View
What: Status and Metrics
Which: Food
How: View Fixed Adding

Alert on drinking tap water in a foreign country if unsafe (P7)

Why: Tracking Elements in View
What: Checks and Alerts
Which: Food and Drink
How: Situated adding

Fig. 9. Examples of ideas from P24 and P7

4.3 Which?

Which (Table 3) refers to the parts of the real world that AR operates on. The previous *What* section focused on the augmentation part of AR and this section describes the reality being augmented. While the most commonly augmented parts of reality are objects and people, this part of the framework highlights what we usually do not think about as targets for augmentation. Beyond just changing people, other living beings, such as animals and plants, can also be augmented, such as making a cat cuter by having it play a virtual flute (T39). Though AR does not change the entire environment like virtual reality, environments can benefit from augmentation. For example, hiding the ceiling of a room to show a starry sky (L121). Adding virtual content can even change the perception of entire activities and events, such as seeing a dead grandparent in the audience during a graduation ceremony (P11). This component of the framework also exposes additional parts of reality we may not think of augmenting, like waste which can be hidden from view when not having time to clean (P12).

4.4 How?

The *What* and *Which* sections above described the different types of AR content and parts of reality that AR operates on. This section section describe how those two are related. The framework categorizes these relationships into three types: additive, representational, and transformative (Figure 10).

4.4.1 Additive. relationships are when AR content is inserted into the real world. In **situated adding**, the AR content is added to the real world but does not appear to be a part of the component of reality that is being augmented. This could be, for example, labels floating next to or above different opaque containers to denote what

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Table 3. Which aspects of reality are useful to augment?

Living Beings	Environments	Actions	Objects
User	Rooms	Activities	Clothes
Plants	Public Spaces	Events	Food
People	Scenery		Waste
Animals	Entire View		Machines
			Media
			Tools
			Vehicles
			Other Objects

they contain (Y164). In contrast, **embedded adding** consists of adding AR content that is spatially registered and appears to be a part of reality being augmented, such as a virtual poster attached to a door that always displays the weather conditions outside (Y129). Lastly, **view-fixed adding** consists of anchoring AR content to the user's view and not to any physical object or location. For example, P13 suggests having the time left before they need to leave always in their view so that they can speed up or relax as needed when getting ready to leave for work.

4.4.2 Representational. relationships consist of AR content that stands in for or mimics real-world entities. This can occur as **copying**, when AR content mimics the appearance or behavior of physical entities, such as a hologram of a friend appearing in the seat beside you while you are traveling (P15), or as **replacing**, which involves AR content acting in place of a physical entity, such as having a virtual Buddha statue in your home instead of a real one, as unlike a real one, it is re-sizable, unbreakable, and not heavy (L42). Another form of representational relationship is **contrasting**, where AR content is deliberately overlaid on corresponding real-world entities to enable direct comparison. This highlights differences between the virtual and physical, such as overlaying a 3D blueprint onto a building under construction (L30) to assess construction accuracy or visualize planned versus actual outcomes.

4.4.3 Transformative. relationships involve AR content altering how physical entities are perceived. This can involve **changing** them, such as making a homemade coffee appear to have fancy latte art on it (P7). **Extending** the dimensions or features of physical entities is also possible, to make them appear larger, more complex, or more interactive than they are in reality. This creates a physical-virtual hybrid entity, for example, a package of gum that has a virtual extension of its packaging containing details about the product that cannot be included in the physical packaging due to size constraints (Y156). Conversely, **diminishing** physical entities is a possibility through AR content that removes or obscures them, with P14 suggesting they wanted to see through the fridge door so they don't have to open it when wanting to know if they have any snacks.

We note that these categories we identify are not mutually exclusive, as AR content may relate to the real world in more than one way at a time. The above example of having a hologram of a friend when traveling is both embedded adding and copying. Similarly, having the time in your view while getting ready is view-fixed adding but also serves as replacing the use of a physical clock.

5 Discussion

Our work describes ways in which AR might integrate into everyday life. We advance the design of future AR experiences by describing the reasons users might want to augment their everyday life (why), the content that is missing from everyday life that AR offers (what), the aspects of everyday life that are useful to augment (which), and how AR content can meaningfully relate to everyday contexts (how). Below, we discuss the implications of our work in terms of how the framework could be used, the differences between the two sources of data

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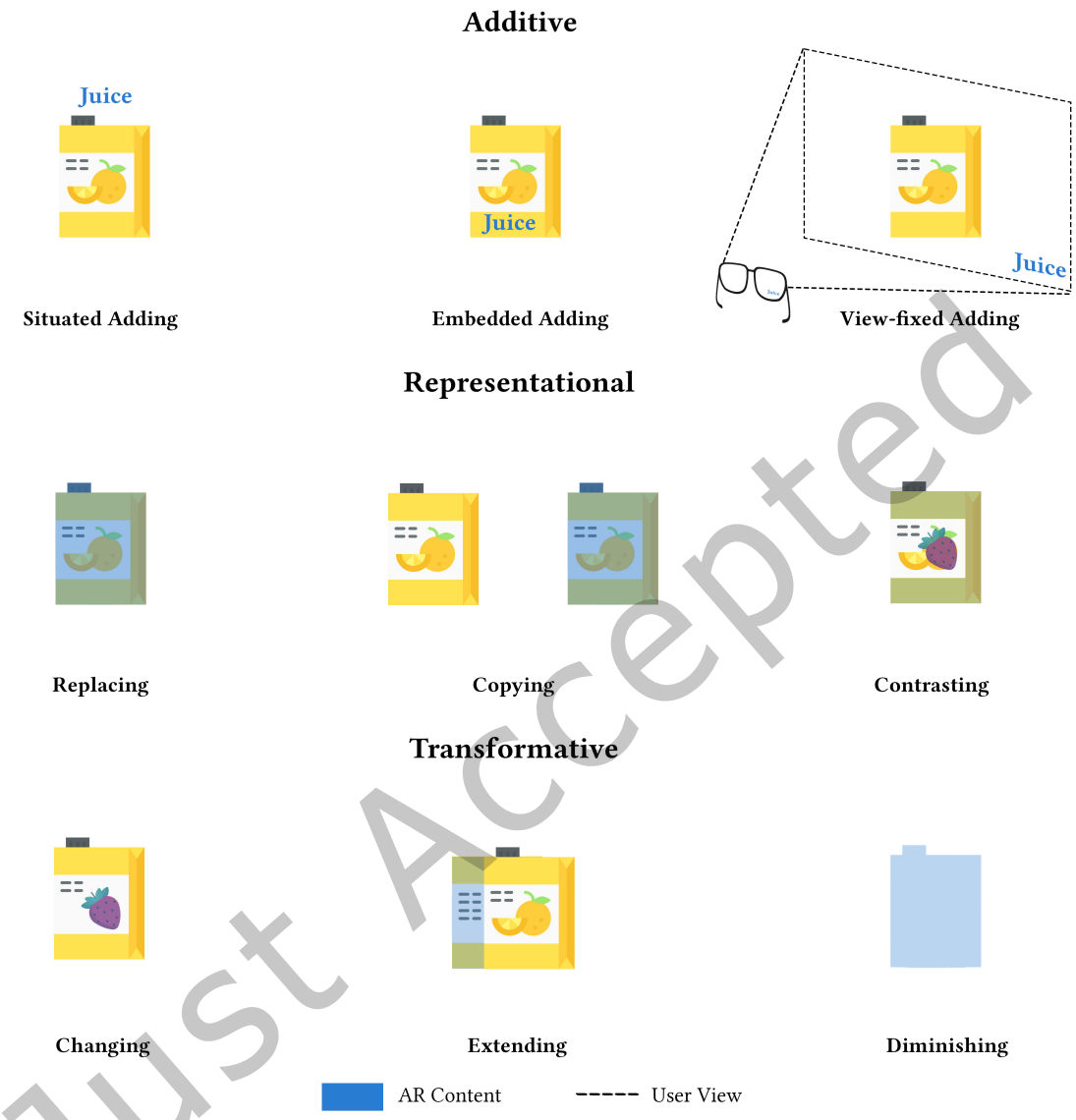

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Fig. 10. The ways AR content can relate to the real world demonstrated using a juice box. In an additive relationship (top row), the text "Juice" can appear floating on top of the box (situated), be anchored to the surface of the box as if a part of it (embedded), or always appear in the corner of the viewer's eye (view-fixed). In a representational relationship (middle row), a virtual juice box may act in place of a real one (replacing), appear next to a real one looking exactly like it (copying), or perfectly aligned and anchored to a real one to show the differences between the real and virtual (contrasting). In a transformative relationship (bottom row), AR content can be added onto the surface of a real juice box to make its packaging look different (changing), an extra AR volume could be added to the real juice box to make it look larger than it is (extending), or the real juice box could be completely hidden from the view by AR content that makes the box look like the background of the scene (diminishing).

we analyzed, what potential harms of everyday AR our work highlights, and the possible impact the usage of everyday AR could have on the real world itself.

5.1 Framework Usage

For AR researchers, the framework provides a structure through which to think about and analyze everyday AR, revealing underexplored areas of research. One such area is augmenting actions, which is highlighted in the *Which* component of the framework. When thinking about AR, we typically focus on augmenting people or objects, but augmenting actions seems overlooked. This oversight may be exemplified by existing research, such as AiGet [24], which allows users to learn about their surroundings through AR during everyday activities. While AiGet covers living beings, environments, and objects, it omits actions, such as dance performances, that could be equally informative and engaging for the user. It is also apparent in AR translation tools, which commonly translate text but rarely address cultural nuances in people's actions (e.g., tipping being considered rude in Japan while not tipping being considered rude in the US), despite their importance in navigating a new environment. Another potential area is proxy objects, which are a part of the *What* and *How* components as *Virtual Counterparts* and *Representational Relationship*. Beyond just acting as a replacement for real objects, future research could reveal new purposes for them, such as manipulating proxy objects locally to affect distant physical counterparts (e.g., preheating the oven before reaching home). Developing ways of supporting specific *Whys* in the framework could also point to further research directions. For example, one could imagine new impossible objects for supporting everyday tasks that can only exist in AR or come up with ways of improving accessibility by moving or transforming physical information for specific users.

For AR designers and developers, the framework highlights user goals for everyday AR systems and design opportunities. It has generative power, such that new ideas for AR use can be devised by combining elements across the four parts of the framework. To demonstrate this, consider a common everyday activity where AR has been extensively applied: driving. The *Why* in AR systems for cars is usually *Access to information in situ*, *Aid Perception*, and *Deictic*. By going through the list of *Why*s in the framework, further applications can be envisioned. Road trips with a virtual friend could be possible by considering *Access to people who are not there*. By considering *Access to physical spaces one doesn't have*, monotonous scenery during a long trip could be changed to a more interesting one to entertain oneself or the passengers, help stay awake, and keep focus on the road.

As everyday AR increasingly relies on AI not just for spatial tracking but for enabling intelligent interactions with the real world [30, 94], the framework can serve as a resource for AR-AI systems and their users. The framework provides a structure for such systems to identify relevant moments during which a user might need assistance, what kind of assistance is appropriate, and ways that assistance could be delivered, something which is underdeveloped in the existing AI-for-AR literature. For example, PARSE-Ego4D [2] uses an LLM to generate action recommendation suggestions for an AR-AI assistant. However, the kinds of actions that the assistant could perform are limited to only six types (search, assistant search, assistant local, language, directions, assistant guide). Our framework could enhance this as we reveal a much larger set of actions possible for AR through the *Why* component of the framework. For users, the framework reveals possibilities for augmenting the world that they might not think of themselves. It also empowers users to articulate their needs more effectively by offering a shared vocabulary for describing their goals and expectations, thereby reducing ambiguity and improving communication between the system and the user.

5.2 Tensions in Practitioner and User Visions

Our analysis of AR visions highlights differences between how users and practitioners imagine technological futures. The most obvious difference is that visions from companies focus on generating commercial value rather than value for users [35]. This is evident in using AR for advertising, which is prominent in the vision

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videos but unsurprisingly absent from the ideas generated by participants in phase 2. Another difference is that practitioner visions focus on generalizability [11, 17]. They aim to create use cases that appeal to a wide audience, rather than trying to target problems faced by smaller groups of people. An example of this is P11's desire to view their late grandparent in the audience during a graduation ceremony, a divisive use case that may be regarded as wholesome or creepy depending on the individual. A related difference to being driven by profit and reputation is that practitioner visions are risk averse. Users instead may be willing to explore negative experiences if they believe these will lead to better outcomes in the long run according to their values, such as P19 wanting to be screamed at by a virtual character on sorting waste incorrectly. Conversely, users may completely ignore commercial and logistical feasibility by proposing one-off use cases that require significant investments in technological infrastructure that are unlikely to occur, such as P11's desire to know exactly how much time it would take to clean their room.

When demonstrating a vision of the future, companies and artists usually prefer to focus on the spectacle of future technologies [31] rather than aspects of life that are practical but rather unexciting. Compared to the scenarios present in the videos from phase 1, the phase 2 scenarios highlight less glamorous aspects of life. This includes personal care (e.g., alert on excessive hair loss), budgeting (e.g., show monthly budget compared to items in the cart during shopping), and filthier examples (e.g., picking up dog poop). Conversely, focusing on spectacle may illuminate possibilities that are not yet a part of everyday experience or knowledge. This can reveal possibilities users might not imagine for themselves, such as impossible objects like Christmas lights that do not respond to gravity (L139). A contrast in the environments where AR usage occurs is also visible across the two phases. Phase 1 shows clean, aesthetically pleasing spaces where AR can work perfectly while phase 2 also shows cluttered spaces with potential distractions and interruptions where AR must work despite imperfections. This difference also extends to how activities are portrayed temporally. Phase 1 shows tasks with clear beginnings and ends, while phase 2 captures activities that unfold as part of ongoing routines amidst multitasking.

A constraint that participant visions suffered in our study is that since they were based on existing tasks and practices, they focused on complimenting what was already happening. Practitioner visions often look beyond this to create completely new workflows such as Samsung re-imaging many aspects of driving in video Y168. Another difference, which we discuss in more detail next, is imagining not just the benefits but also the harms of future technologies. Examples of misuse of technology, such as tracking staff in a store to know when it is safe to steal (P25) are present in visions from participants but completely absent from visions from companies.

Taken together, these contrasts illustrate how the two phases provide complementary strengths. Phase 1 contributes breadth, expertise, and long-range imaginaries while study 2 contributes depth, realism, and situatedness. Thus, the combination of our two sources enables a more comprehensive and balanced understanding of how AR may integrate into everyday life than either source alone.

5.3 Potential Harms of Everyday AR

By making technology even more ubiquitous than it already is, everyday AR risks amplifying existing technological harms and introducing new ones. Participants in our study suggested not just to extend their cognitive processes through AR but often to completely offload them. They wanted AR systems to be aware in their stead for tasks such as biking (P2), packing before leaving the house (P17), and noticing if they are coughing too much (P13). While such use cases would make life easier for users, doing so would heighten the existing cognitive harms caused by technology such as digital dementia [93], skill erosion [84], and complacency [39]. Thus, there is a need to study and discuss which cognitive functions should be offloaded to AR and to what extent. Just as automobiles have speed limits to protect users, everyday AR may require safeguards that hold back users from the full capabilities of the technology.

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Besides offloading cognition, participants were willing to give up agency to AR systems, such as for deciding how to place things in the kitchen (P18), which checkout line to take in the grocery store (P13), what order to load the dishwasher in (P4), and if they should take the stairs or a lift (P13). These micro-optimizations that offload decision making might save a couple of minutes for the user but can quickly turn annoying or cause diminished autonomy as users act mindlessly because they think that the AR systems knows what is best for them [10, 68]. Yet, in high-stress or unfamiliar situations, such delegation could be beneficial. Determining how much AR systems should take over and under what conditions remains an open research challenge.

Some participants even suggested offloading responsibility for making them do certain actions to AR systems. They wanted to be nagged (P19), shown potential consequences for their actions (P5), and even shocked (P18) into doing what they considered right but could not bring themselves to do on their own. Using AR for behavior change is an exciting research area but these interventions need to be designed to foster sustainable habits, rather than to act as an excuse for bad behavior in the absence of the system or a crutch without which users are unable to do what they deem is right [28].

Potential issues with social AR also surfaced in our study. P12 wanted AR to create a virtual referee for their badminton game with their friend, preferring a fully embodied AR figure over simple text-based rule alerts. This choice raises concerns about how information should be conveyed in social AR settings and the implications of avatar embodiment [47, 99]. The presence of a virtual person may enhance immersion and clarity but could also introduce discomfort or distraction depending on the context and social norms. Moreover, P5 desired to use AR to track when household items were last cleaned to increase accountability and transparency in shared living spaces. While this may improve hygiene and reduce conflict, it could also introduce surveillance-like dynamics and echoes with previous research on the fine line between helpfulness and coercion [76]. The concerns raised by shared living spaces also point to a broader question about the ownership of spaces, objects, and even the self, and whether people have the right to change what others perceive. Neely [74] frames this issue by distinguishing between public and private spaces, arguing that augmenting private spaces is analogous to modifying private property without permission, while leaving open the question of whether and under what conditions should public spaces be augmentable. Extending this discussion, Wolf et al. [102] argue that AR requires clear notions of ownership over virtual spaces, including who has the authority to determine what augmentations are permitted within them. They further contend that individuals should retain ownership over their own bodies and control over how they are represented and perceived in AR.

Collectively, these examples illustrate that everyday AR is vulnerable to dark patterns and unintended consequences. Care must be taken to ensure that these systems empower users rather than erode cognitive skills, autonomy, or social trust.

5.4 Impact on the Real World

While AR can digitally augment the world by overlaying virtual content, it may also transform the actual physical world in the long term. According to activity theory [52, 98], human activity is mediated by tools, which shape not only how tasks are performed but also how individuals perceive and engage with their environments. AR, as a mediating tool, can both enhance existing activities and transform which activities are performed and how. Similar shifts have occurred with other technologies, such as desktop computers altering the role of paper in offices [19, 89].

In our study, P10 mentioned wanting to use AR to hide wrinkles in their clothes to never again need to iron their clothes. Taking this a step further, another participant expressed their desire to wear pajamas all the time to feel comfortable but appear to be wearing a gown to others. These scenarios suggest that AR could alter the symbolic function of fashion, liberating individuals from social norms regarding appearance, thus allowing for greater personal comfort and expression.

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The ability to alter appearances also raises a broader question: When all visual appearance can be digitally manipulated by AR, what becomes of objects whose primary function is aesthetic? Moving from clothes to hair, one participant wanted to try out hairstyles with AR before getting a haircut. If AR can simulate hair in real-time, it could save time, money, and resources, potentially reducing the demand for hair products and services. On the flip side, it could also take a dystopian turn with hair becoming a luxury leading to AR causing novel forms of social stratification, as forewarned by O'Hagan et al. [76].

From an activity theory perspective, the tool (AR) transforms the object (e.g., clothing, hair, furniture) not just in appearance but in its social and functional meaning. This mediation could lead to a redefinition of what is considered real, necessary, or valuable in everyday life. Exploring the impact that ubiquitous AR may have on various objects and the implications of this impact are important directions for future research on everyday AR. In particular, research could investigate not only user preferences for AR or physical objects but also social norms, market shifts, and ecological consequences arising from a reduced need for physical objects.

6 Limitations and Future Work

Our work analyzes how people imagine a future with AR to create a framework for how AR would integrate with everyday life. The framework is grounded in a finite set of visions, some of which assume capabilities that current AR systems do not yet possess, and hence, may not reflect how such a future is ultimately realized, as everyday life in the future is shaped by various factors such as emerging technologies, practical considerations, and evolving cultures. It is meant to act as a snapshot of how we currently imagine a future with ubiquitous AR and is valuable for guiding the development of future AR applications and helping users understand what AR can do for them.

Through our study, we were able to capture scenarios representing everyday life, including situations such as socializing with friends, riding at full speed on a bike, and even one participant kissing their partner. However, some aspects of life remain difficult to access through self-report or observation, such as using the toilet, sexual activities, or having an argument. Engaging with such private or taboo aspects of life has been identified as a challenges for HCI [12, 42] and requires study designs that explicitly prioritize sensitivity and privacy, something future work could explore more deliberately

Regarding our study design, asking users what they want may be characterized as a limited method [63], as users do not always possess the insights needed to produce truly innovative inventions. Yet, user-centered design must balance creating systems that do not yet exist with a grounding in existing, complex practices. In this work, we have sought to have users participate by combining speculative visions with reflections on their everyday practices. In this way, we connect back to earlier HCI work [3, 41] where diary methods, for example, were used in designing early smartphone applications [91], offering insights into the smartphone-enabled practices that followed.

Further, our study represents only an initial step toward understanding AR's integration into the broad and heterogeneous domain of everyday life, which varies significantly across individuals, locations, and values. The participants in our study are predominantly young individuals living in a developed Western country, which limits the generalizability of our framework. As adoption of AR increases, future work will need to examine AR use across more diverse age groups, including parents, children, and older adults. Another aspect of everyday life that future studies could explore is AR use in other parts of the world. We acknowledge that the participants in our study were WEIRD (Western, Educated, Industrialized, Rich, and Democratic) and thus our framework may have limited applicability in non-Western contexts. Participants being WEIRD is a known problem with research at CHI and a potential solution proposed by Linxen et al. [60] is to replicate studies across geographic locations. Thus, future work could apply our methodology in different parts of the world to uncover new possibilities for AR and also compare how differences such as value or time or availability of infrastructure impacts AR use. In

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the past, research on how smartphones integrate into everyday life followed a similar trajectory where initial studies focused on general users [20, 21] and were later complemented by studies of parents [100], older adults [85], and individuals from non-Western countries [38].

7 Conclusion

We have mapped out the "Promise of Augmented Reality in Everyday Life" to explain how AR could extend computing beyond screens and integrate into our everyday lives. We do this by analyzing visions of how AR could be used in the future collected from two complementary sources: (1) vision videos created by companies or professionals, and (2) visions generated by adding AR content to first person recordings of people's everyday lives. The framework resulting from our analysis providing a structure to understand and ideate about everyday AR use. At a time when AR capabilities are improving and adoption increases, our work moves AR one step closer to becoming the next personal computing device by highlighting the role it could play in our everyday lives.

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A Participant Demographics

Table 4. Participant Demographics for Study 2

PID	Age	Gender	Country of Origin	Occupation Type	Living Situation
P1	27	Male	Denmark	Office Job	Living Alone
P2	23	Female	Denmark	Student	Student Dormitory
P3	28	Male	Denmark	Student	Student Dormitory
P4	28	Male	India	Education or Academia	Shared Apartment
P5	34	Female	Poland	Office Job	Shared Apartment
P6	31	Male	Denmark	Remote Work	Shared Apartment
P7	27	Female	India	Office Job	Shared Apartment
P8	30	Female	India	Public Service	Shared Apartment
P9	28	Male	India	Remote Work	Cohabiting with Partner
P10	26	Female	Italy	Student	Student Dormitory
P11	39	Female	Canada	Education or Academia	Cohabiting with Partner
P12	26	Female	India	Education or Academia	Living with Family
P13	25	Female	Germany	Education or Academia	Shared Apartment
P14	25	Female	Denmark	Student	Shared Apartment
P15	35	Female	Germany	Education or Academia	Cohabiting with Partner
P16	23	Male	Portugal	Student	Student Dormitory
P17	26	Female	Poland	Student	Student Dormitory
P18	29	Male	Italy	Office Job	Shared Apartment
P19	26	Female	India	Unemployed	Shared Apartment
P20	27	Male	Denmark	Public Service	Cohabiting with Partner
P21	27	Female	Thailand	Service Job	Cohabiting with Partner
P22	21	Female	Czech Republic	Student	Student Dormitory
P23	30	Female	Denmark	Office Job	Living Alone
P24	35	Male	Denmark	Creative Professional	Cohabiting with Partner
P25	30	Male	Denmark	Office Job	Cohabiting with Partner
P26	27	Female	Spain	Office Job	Cohabiting with Partner
P27	23	Male	Portugal	Student	Student Dormitory
P28	21	Female	Denmark	Student	Living with Family
P29	33	Female	Poland	Creative Professional	Shared Apartment
P30	25	Female	Denmark	Creative Professional	Living Alone

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